

AI-Powered Multimodal Contactless Patient Monitoring & Clinical Decision Support System

Using the Hospital Bed as an Unobtrusive Sensing Platform

1. Problem Statement

Background

Continuous monitoring of hospitalized patients is essential for early detection of clinical deterioration. Today, hospitals primarily rely on contact-based medical devices such as ECG electrodes, pulse oximeters, blood pressure cuffs, respiratory belts and temperature probes.

Although these devices provide accurate measurements, they have several practical limitations:

- Multiple sensors must be attached to the patient's body.
- Sensors may become detached due to movement.
- Long-term attachment can cause discomfort.
- Nurses must frequently reposition or maintain sensors.
- Different devices operate independently, requiring healthcare staff to interpret multiple streams of information.

The challenge is not simply obtaining physiological measurements—it is obtaining them continuously, comfortably and intelligently.

2. Vision

Our objective is **not to build another smart hospital bed**.

Our objective is to develop a **contactless patient monitoring platform** capable of continuously observing patients without attaching sensors to their body.

The hospital bed serves only as an unobtrusive sensing platform because every admitted patient naturally spends a significant amount of time on it.

Artificial Intelligence then transforms these observations into clinically meaningful information that assists doctors and nurses in decision-making.

3. Existing Monitoring Methods

Contact-Based Monitoring

Current hospitals commonly use:

- ECG Electrodes
- Pulse Oximeters
- Blood Pressure Cuffs
- Temperature Probes
- Respiratory Belts

Advantages

- High accuracy
- Clinically validated

Limitations

- Physical attachment required
 - Patient discomfort
 - Frequent maintenance
 - Sensor displacement
 - Multiple independent devices
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Contactless Monitoring

Some modern hospitals also employ smart beds capable of monitoring:

- Heart Rate
- Respiratory Rate

- Movement
- Bed Occupancy
- Pressure Distribution
- Fall Risk

However, these systems are:

- Expensive
 - Limited to premium hospitals
 - Primarily monitoring devices
 - Not designed as intelligent clinical assistants
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4. Research Gap

Current monitoring systems successfully collect patient data.

However, several important challenges remain.

Contact Dependency

Many physiological parameters still require sensors attached directly to the patient.

Fragmented Information

Different devices generate different measurements.

Healthcare professionals must combine this information mentally before making decisions.

Limited Intelligence

Most systems answer:

What happened?

Few systems answer:

- Why did it happen?
 - How reliable is the observation?
 - Which patient needs attention first?
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Practical Deployment

Real hospital environments contain:

- Visitors sitting beside patients
- Patient movement
- Nurses repositioning patients
- Environmental interference

These factors reduce measurement reliability but are rarely considered explicitly.

5. Proposed Solution

We propose an **AI-powered multimodal contactless patient monitoring platform**.

Instead of attaching multiple sensors to the patient, the system integrates unobtrusive sensors into the hospital bed and surrounding bed structure.

These sensors continuously observe the patient's physiological and behavioural characteristics.

Artificial Intelligence combines information from multiple sensors, validates signal reliability, and converts raw observations into clinically useful insights.

The goal is not to replace existing medical equipment immediately, but to reduce dependence on contact-based monitoring while providing intelligent clinical decision support.

6. Clinical Information Required

Doctors require information—not raw sensor data.

The system aims to estimate the following clinical parameters.

Clinical Information	Clinical Importance
Heart Rate	Cardiac status
Respiratory Rate	Respiratory distress
Body Temperature	Fever / Infection
Body Movement	Recovery / Restlessness
Body Position	Pressure ulcer prevention
Pressure Distribution	Bedsore risk
Bed Occupancy	Patient safety
Weight	Fluid retention
Sleep Behaviour	Recovery quality

7. Why Use the Hospital Bed?

The hospital bed is not the product.

It is the sensing platform.

Advantages include:

- Every admitted patient already uses it.
- Sensors remain completely hidden.
- No behavioural changes are required from the patient.
- Continuous monitoring becomes possible without wearable devices.
- Existing beds can potentially be retrofitted.

8. Sensor Architecture

The system uses multiple complementary sensors.

Each sensor observes a different aspect of the patient's condition.

No single sensor is trusted independently.

9. Sensor Placement & Working Principle

9.1 mmWave Radar

Placement

Mounted inside the headboard and directed toward the patient's chest.

Principle

The radar emits low-power millimeter-wave signals.

As the chest expands and contracts during breathing, the reflected waves change.

Tiny chest movements can also reveal heartbeat-related micro-motion.

Measures

- Respiratory Rate
- Chest Motion
- Heartbeat Estimation

Advantages

- Completely contactless
- Works under blankets
- Continuous monitoring

Limitations

- Large body movement reduces accuracy.
- Nearby people may interfere if not properly positioned.

AI Interpretation

The AI first checks whether:

- The patient is relatively still.
- Only one occupant is present.

Only then is radar data considered reliable.

9.2 Pressure Sensor Matrix

Placement

Embedded beneath the mattress.

Principle

Hundreds of pressure sensing elements continuously measure pressure distribution across the mattress.

Measures

- Body Position
- Pressure Distribution
- Movement
- Bed Occupancy

Advantages

- Large sensing area
- Reliable posture estimation
- Pressure ulcer monitoring

Limitations

- Cannot directly measure physiological signals.

AI Interpretation

AI detects:

- Reduced movement
 - Frequent turning
 - Prolonged pressure
 - Bed exit attempts
-

9.3 Load Cells

Placement

Installed beneath each leg of the bed.

Principle

Continuously measures the total weight acting on the bed.

Measures

- Patient Weight
- Bed Occupancy
- Additional Occupants
- Bed Exit

Advantages

- Simple
- Reliable
- Low Cost

Limitations

Cannot identify who is sitting on the bed.

AI Interpretation

If weight suddenly increases while pressure distribution changes,

AI recognises:

Visitor present

and reduces confidence in other physiological measurements.

9.4 Thermal Camera

Placement

Mounted above the patient's upper body.

Principle

Measures emitted infrared radiation.

Produces a thermal map instead of a colour image.

Measures

- Skin Temperature
- Temperature Distribution

Advantages

- Contactless
- Continuous

Limitations

- Blankets reduce accuracy.
- Measures surface temperature only.

AI Interpretation

Temperature trends are analysed together with respiration and movement.

10. Sensor Fusion Strategy

Instead of relying on one sensor, the AI validates information across multiple sensors.

Example:

Pressure Matrix

↓

Patient is still

↓

Load Cells

↓

Single Occupant Confirmed

↓

mmWave Radar



Respiration Reliable



Thermal Camera



Temperature Reliable



Sensor Fusion Engine



Clinical Interpretation

This approach improves reliability while reducing false alarms.

11. AI Pipeline

The AI performs six primary functions.

Signal Processing

Removes sensor noise.

Sensor Validation

Determines whether each measurement is reliable.

Sensor Fusion

Combines multiple sensor observations.

Clinical Interpretation

Converts observations into clinically meaningful information.

Patient Prioritization

Ranks patients according to urgency.

Explainability

Provides the reason behind every prediction.

Example:

High Priority

Reason:

- Respiration increasing
- Reduced movement
- Elevated temperature

Confidence:

92%

12. Expected Dashboard

Instead of displaying isolated measurements,
the dashboard displays actionable information.

Example

Patient ID

12

Priority

High

Reason

- Respiratory deterioration suspected
- Pressure ulcer risk increasing

Confidence

94%

Suggested Action

Review patient within 15 minutes.

13. Hardware Components

Sensors

- mmWave Radar
- Pressure Sensor Matrix
- Load Cells
- Thermal Camera

Processing

- ESP32
 - Raspberry Pi 5
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14. Software Components

Programming

- Python
- C++

AI

- PyTorch

- Scikit-learn

Backend

- FastAPI

Communication

- MQTT

Database

- PostgreSQL / SQLite

15. Estimated Prototype Cost

Component	Approximate Cost
Raspberry Pi 5	₹8,000
ESP32	₹700
mmWave Radar	₹18,000
Pressure Sensor Matrix	₹10,000
Load Cells	₹3,000
Thermal Camera	₹6,000
Electronics & Fabrication	₹10,000

Estimated Total

₹50,000–70,000

16. Technical Feasibility

Area	Rating
Hardware	9/10

Software	9/10
AI	9/10
Sensor Fusion	9/10
Prototype	9/10
Clinical Validation	5/10

Overall Feasibility

9/10

17. Challenges

Technical

- Sensor synchronization
- Environmental interference
- Patient movement
- Calibration
- Signal reliability

Clinical

- Validation against medical equipment
- Ethical approval
- Clinical testing

Deployment

- Cost optimization
- Scalability
- Hospital integration

18. Limitations

The proposed system currently cannot replace:

- ECG
- Blood Pressure Monitor
- Pulse Oximeter (SpO₂)

These measurements still require direct-contact or clinically validated technologies.

The platform should therefore be viewed as an intelligent observational and clinical decision-support system rather than a replacement for all hospital monitoring equipment.

19. Research Contributions

This work contributes:

- A low-cost multimodal contactless monitoring platform.
 - Use of the hospital bed as an unobtrusive sensing foundation.
 - AI-based sensor fusion for reliable contactless monitoring.
 - Confidence-aware measurement validation.
 - Explainable clinical decision support.
 - Patient prioritization based on multimodal observations.
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20. Future Scope

- Contactless Blood Pressure Estimation
- Contactless SpO₂ Estimation
- Personalized Patient Digital Twins
- Electronic Health Record Integration
- Ward-Level AI Coordination
- Multi-Hospital Federated Learning
- Predictive Clinical Deterioration Analytics